

# PAPER\_681: Gross-Pitaevskii Vortex Simulation of the UQFF Aether Around Black Holes

**Author:** Daniel Murphy  
**Framework:** Unified Quantum Field Framework (UQFF) v5.30  
**Session:** 173  
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**Class:** GrossPitaevskiiVortexSimulation | CP4 #265  
**Source:** grok\_share\_fc21e30c24b4.txt  
**Domain:** BEC / Quantum Gravity Interface

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## Abstract

We numerically solve the radial Gross-Pitaevskii equation for the [UA] Aether wavefunction in the gravitational potential of a black hole, incorporating the UQFF magnetic string term  $U_m(r, t)$ . Using imaginary-time propagation we obtain the ground-state density profile  $|\psi(r)|^2$ , chemical potential  $\mu_{\text{UA}}$ , and demonstrate aether density enhancement  $n_{\text{UA}}(r) = N_{\text{UA}}(1 + r_s/r \cdot f_{\text{TRZ}})$  near the horizon.

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## Primary UQFF Equation

$$i\hbar \frac{\partial \psi}{\partial t} = \left[ -\frac{\hbar^2}{2m_{\text{UA}}} \nabla^2 + V_{\text{grav}}(r) + g_{\text{UA}}|\psi|^2 + U_m(r, t) \right] \psi$$

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## Parameters

| Parameter         | Value                             | Description        |
|-------------------|-----------------------------------|--------------------|
| $M$               | $8.548 \times 10^{36} \text{ kg}$ | Sgr A* mass        |
| $r_{\text{min}}$  | $r_s$                             | Inner boundary     |
| $N_{\text{grid}}$ | 100                               | Radial grid points |

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## UQFF Constants

| Constant            | Value                                 | Description              |
|---------------------|---------------------------------------|--------------------------|
| $\rho_{\text{UA}}$  | $7.09 \times 10^{-36} \text{ kg/m}^3$ | Universal Aether density |
| $\rho_{\text{SCm}}$ | $7.09 \times 10^{-37} \text{ kg/m}^3$ | Schwarzschild condensate |

| Constant         | Value                                        | Description                  |
|------------------|----------------------------------------------|------------------------------|
| $f_{\text{TRZ}}$ | 0.1                                          | Time-reversal zone factor    |
| $\kappa$         | $5 \times 10^{-4} \text{ day}^{-1}$          | UQFF calibration constant    |
| [SSq]            | 0.57                                         | Superstring quenching factor |
| $\mu_J$          | $3.38 \times 10^{23} \text{ J}\cdot\text{m}$ | Magnetic string coupling     |
| $\gamma$         | $5 \times 10^{-5} / 86400 \text{ s}^{-1}$    | Aether oscillation decay     |

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## C++ Implementation

```
#include "GrossPitaevskiiVortexSimulation.h"
```

```
GrossPitaevskiiVortexSimulation obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

## Python CP4 Calculator

```
from CondensedPhysics2 import GrossPitaevskiiVortexSimulationCalculator

calc = GrossPitaevskiiVortexSimulationCalculator()
result = calc.compute() # returns all UQFF quantities
print(result)
```

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## References

Gross 1961; Pitaevskii 1961; Berezhiani & Khoury 2015 (superfluid DM); UQFF Session 173

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